Introduction to Bioinformatics

Single-cell RNA sequencing

Ali Sharifi-Zarchi   Somayyeh Koohi

Department of Computer Engineering, Sharif University of Technology

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Presentation Overview

The building block of life

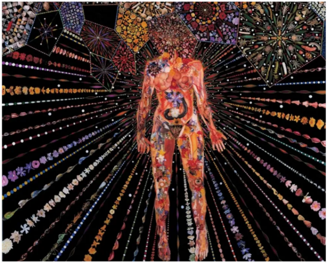
- **Life** distinguishes living from dead or inanimate entities.
- Most definitions of life share a common entity: **cells**.
- Cells form open systems which:
  - maintain homeostasis,
  - have metabolism,
  - grow and adapt,
  - reproduce,
  - respond to stimuli,
  - and organize themselves.

- Therefore, **cells are the fundamental building blocks of life**.
- Cells were first described in 1665 by **Robert Hooke**, who observed a thin slice of cork and named the tiny units “cells”.

The Cell Atlas

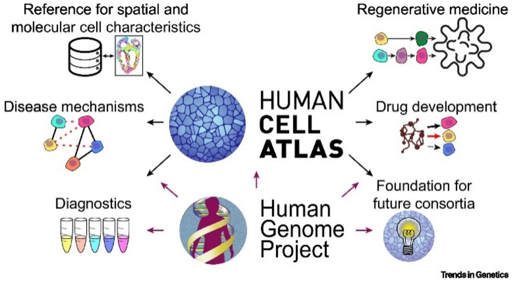
- Biology's next mega-project: individually capture and scrutinize **millions of cells**.
- Objective: build a comprehensive **"cell atlas"** (a map of human cells).

- Promise:
  - reveal what human bodies are made of,
  - provide a sophisticated new model of biology,
  - accelerate the search for drugs.
- Reference: MIT Technology Review — The Cell Atlas.

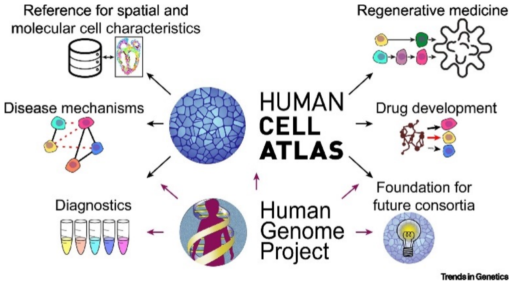


Towards a Human Cell Atlas (HCA)

- Characterizing cellular composition and organization of tissues has been a long-term challenge.
- **Human Cell Atlas (HCA)**: global collaborative effort to create a **reference map of all human cells**.



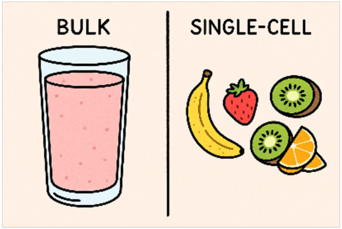
Towards a Human Cell Atlas (HCA)



- Purpose:
  - understand human health,
  - diagnose, monitor, and treat disease.
- Analogous to the **Human Genome Project (HGP)** in ambition and impact.
- Reference: Trends in Genetics (2021).

Single-cell RNA sequencing

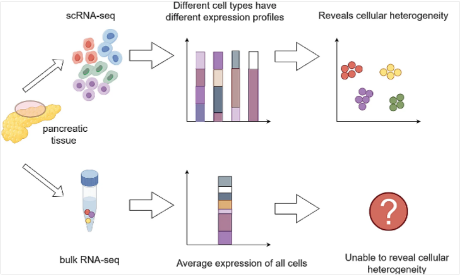
- Sequencing RNA provides snapshots of cells/tissues/organisms as **gene expression profiles**.
- Used to detect changes:
  - across disease states,
  - in response to therapeutics,
  - under different environments,
  - across genotypes / experimental designs.



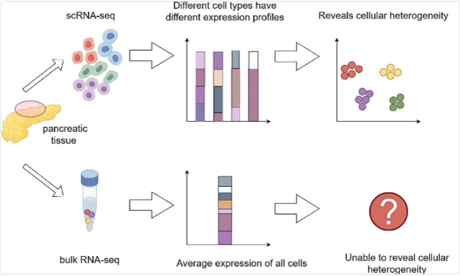
- Two common modes:
  - **Bulk RNA-seq**: mixed RNA from all cells in a sample.
  - **Single-cell RNA-seq (scRNA-seq)**: transcriptomes of **individual** cells.
- Fruit analogy: bulk is a **smoothie**; single-cell keeps each fruit **separate** to measure each "flavor".

Bulk vs scRNA-seq: Why single-cell?

- Bulk RNA-seq is usually **cheaper** and **simpler**.
- But bulk yields **cell-averaged** profiles that hide:
  - expression heterogeneity between cells,
  - rare or transient cell states.
- Some perturbations affect only specific cell types or interactions.



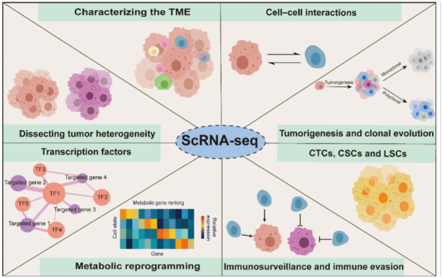
Bulk vs scRNA-seq: Why single-cell?



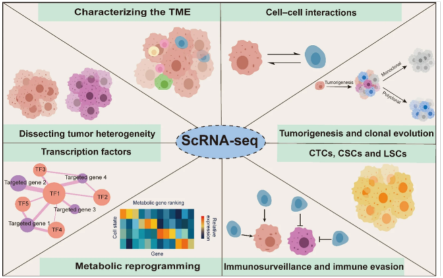
- Example in oncology:
    - **rare drug-resistant tumor cells** can drive relapse,
    - difficult to detect with bulk RNA-seq.
- Trade-off: scRNA-seq is often more expensive/harder and downstream analyses are more complex (risking false conclusions).
- Reference: J Transl Med (2024).

Applications of scRNA-seq in human cancer biology

- Resolves tumor heterogeneity and identifies distinct malignant/non-malignant cell states.
- Profiles the tumor microenvironment (TME):
  - immune diversity,
  - stromal diversity.
- Can infer:
  - cell–cell interactions,
  - TF-driven regulatory networks,
  - metabolic reprogramming signatures.

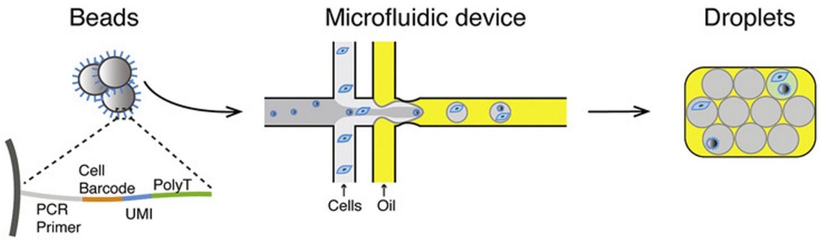


Applications of scRNA-seq in human cancer biology



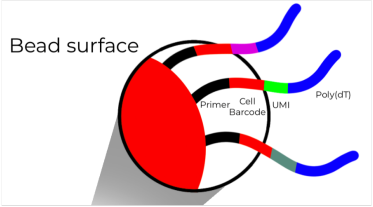
- Supports reconstruction of tumorigenesis and clonal evolution (mono- vs polyclonal patterns).
- Improves characterization of rare or clinically relevant populations:
  - Circulating tumor cells (CTCs), cancer stem cells (CSCs), and leukemia stem cells (LSCs)
  - mechanisms of immunosurveillance and immune evasion.
- Reference: Cell Death & Disease (2023).

Microfluidic device based protocols



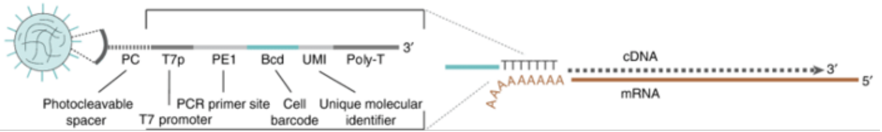
- Microfluidic single-cell strategies trap cells inside hydrogel droplets.
- Droplets act as **single-cell reaction chambers** (compartmentalisation).
- Technology relies on capturing mRNA poly-A tails using poly-dT sequences attached to beads.
- Reference: Molecular Cell (2018).

Barcodes and UMIs



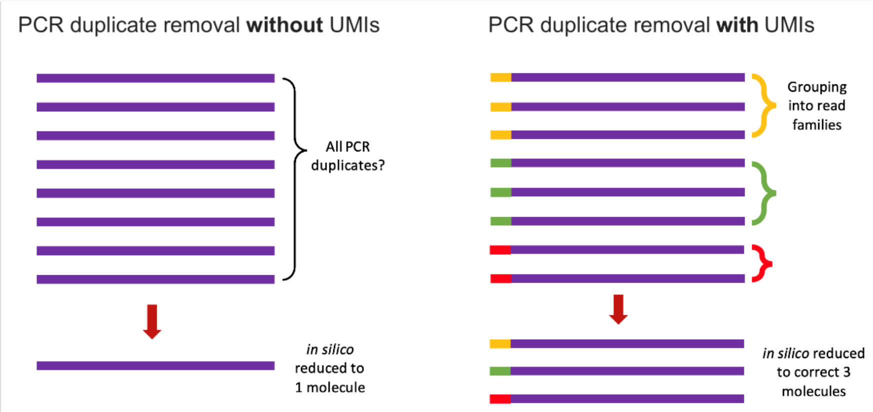
- **Cell Barcodes** are short nucleotide tags that are used to **label** sequences. These labels are used to sort sequences that have originated from a single cell source.
- A **cell barcode** and a **UMI** are crucial for interpreting single-cell libraries and read quality.
- Single-cell technologies often generate **paired-end** reads.
- Structure commonly includes:
  - **Barcode** followed by a **UMI**,
  - located at either the 3′ or 5′ end (platform-dependent).
- Reference: Geneious Biologics help article.

Why UMIs?



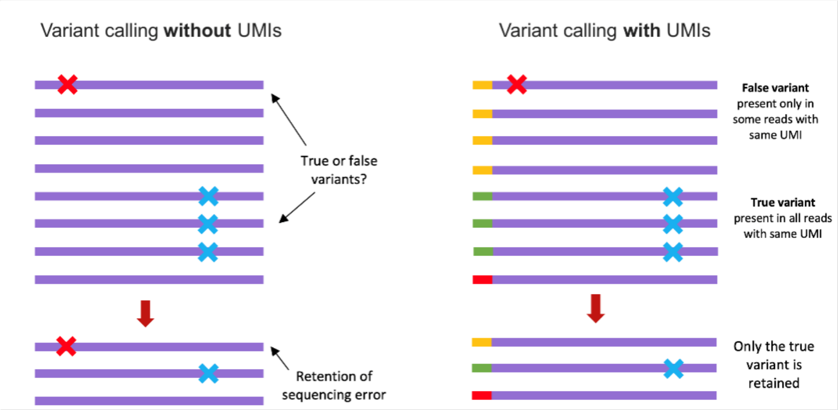
- During transcript amplification (typically PCR), copies are made from identical fragments.
- Copies and original molecules are indistinguishable $\Rightarrow$ hard to infer original molecule counts.
- **UMIs (Unique Molecular Identifiers)** solve this by tagging each original molecule with a short random sequence.
- Practical benefits:
  - estimate starting mRNA content (each UMI $\approx$ one initial mRNA),
  - distinguish true vs false variants.
- Reference: Nature Protocols (2016).

Benefits of UMI



- UMIs can be used to determine the starting mRNA content, as each UMI represents one initial mRNA read.

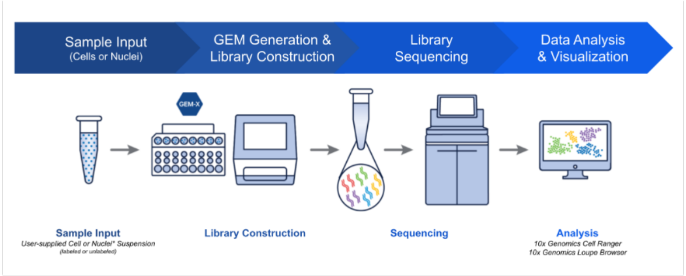
Benefits of UMI



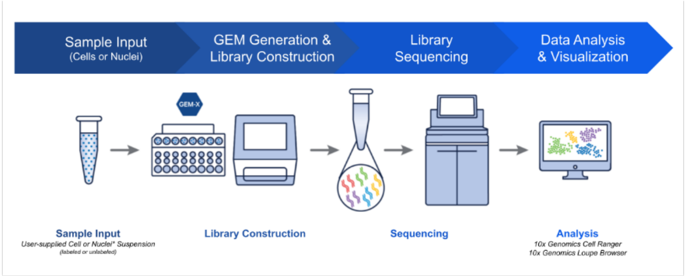
- UMIs can help to distinguish between true and false variants.

Droplet workflow (high-level)

- Encapsulation is designed to capture **beads and cells simultaneously**.
- Beads carry on-bead primers containing:
  - PCR handle,
  - cell barcode,
  - 4–8 bp UMI (typical),
  - poly-T tail (or poly-T primer for 5’ kits).



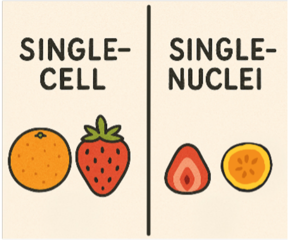
Droplet workflow (high-level)



- After lysis: mRNA released and captured by barcoded oligos on beads.
- Droplets are broken to release STAMPs (single-cell transcriptomes on microparticles).
- Followed by reverse transcription + PCR amplification.
- **Tagmentation**: transcripts are cut and sequencing adapters are attached $\Rightarrow$ ready libraries.
- Reference: An introduction to GEM-X technology.

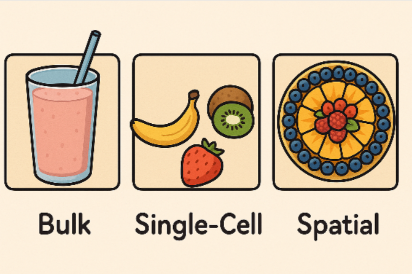
Single-Cell vs Single-Nuclei

- Single-cell profiling can be biased for some tissues (e.g., brain).
- Tissue dissociation can:
  - damage vulnerable cell types,
  - reduce capture of certain populations,
  - require fresh tissue (limits biobank use).



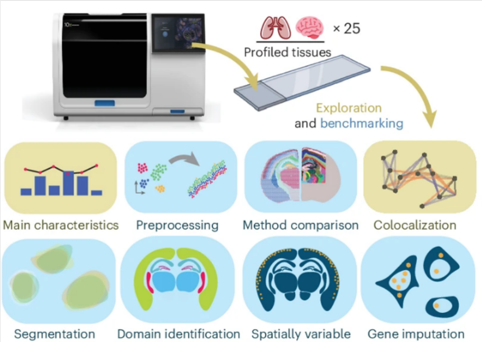
- Nuclei are more resistant and can be isolated from **frozen** tissue without dissociation enzymes.
- Nuclei have been shown to accurately reflect transcriptional patterns.
- Choice depends mainly on **tissue type** and sample constraints.

The Development of Spatial Transcriptomics



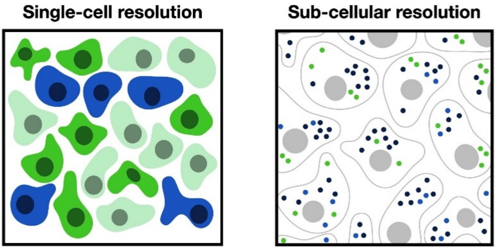
- Standard single-cell technologies require dissociation $\Rightarrow$ spatial context is lost.
- Spatially resolved genomics preserves spatial information while measuring genome-scale omics.
- Fruit tart analogy: each fruit has its own distinct **place** relative to others.

Xenium In Situ platform



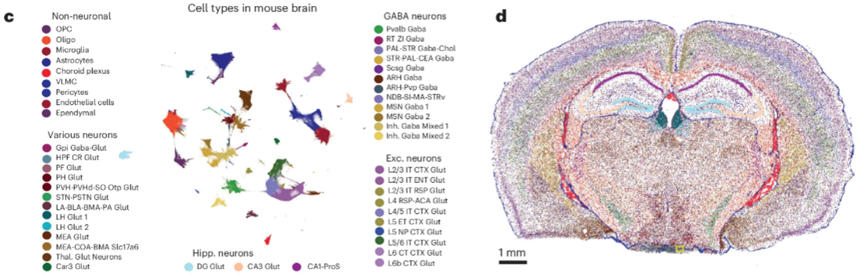
- Imaging-based spatial transcriptomics:
  - targeted and highly multiplexed detection of individual RNA molecules,
  - fluorescence-based microscopy.
- **Xenium** (10x Genomics) claims maps of **hundreds of genes** at **subcellular resolution**.
- Reference: Nature Methods (2025).

From subcellular spots to single cells



- Spatial omics at subcellular resolution captures positions of individual RNA molecules.
- Positions captured by:
  - single-molecule imaging, or
  - spatial barcoding with spot sizes smaller than single cells.
- Cell segmentation on subcellular data enables single-cell resolution:
  - aggregate expression to cell-wise measurements,
  - then process with spatially-aware analysis pipelines.

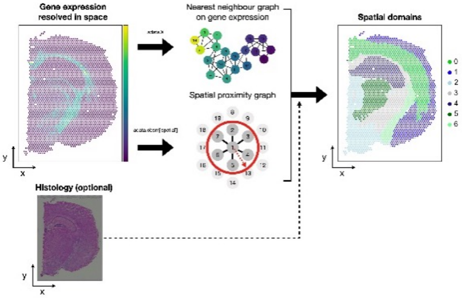
Xenium example visualizations



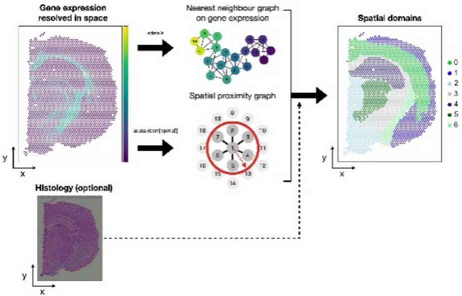
- UMAP of cells in multiple tissue sections, colored by cell type.
- Spatial map of inferred cell types in the tissue.

Spatial domains

- Spatial omics data includes:
  - a cell $\times$ gene matrix,
  - tissue image,
  - spatial coordinates.
- Early analysis task: identify cell types/states to generate hypotheses.



Spatial domains



- Common approach:
  - compute a nearest-neighbor graph on a low-dimensional embedding,
  - apply community detection (clustering).
- Spatial extension: incorporate similarity in **coordinate space** alongside gene-expression similarity.
- This is the **identification of spatial domains**.
- Reference: Single-cell best practices.

References (from slides)

- sc-best-practices.org
- MIT Technology Review — The Cell Atlas
- Towards a Human Cell Atlas (Trends in Genetics, 2021)
- Bulk vs scRNA-seq discussion (J Transl Med, 2024)
- scRNA-seq in cancer biology (Cell Death & Disease, 2023)
- Microfluidic protocols (Molecular Cell, 2018)
- Barcodes and UMIs (Geneious Biologics)
- UMIs (Nature Protocols, 2016)
- Droplet workflow (An introduction to GEM-X technology).
- Xenium In Situ platform (Nature Methods, 2025)
- Spatial domains (Single-cell best practices).

The End

Questions? Comments?

Slides prepared by
Aysan Neghabi and **Mahdi Hassanzadeh**